The following table lists connected networks of each router

Router	Available networks on local interfaces	Network available on other routers interfaces
Router 0	10.0.0.0/8, 20.0.0.0/8, 40.0.0.0/8	30.0.0.0/8, 50.0.0.0/8
Router 1	20.0.0.0/8, 30.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 40.0.0.0/8
Router 2	40.0.0.0/8, 50.0.0.0/8	10.0.0.0/8, 20.0.0.0/8, 30.0.0.0/8

15-9-20

EX-1/ Simulation of Static Routing Configuration using CISCO Packet Tracer

AIM:

To simulate Static Routing Configuration using CISCO Packet Tracer.

PROCEDURE:

Setting up a practice lab

Create a packet tracer lab or download the following pre created lab and load it on Packet Tracer

In this lab each network has two route to reach. We will configure one route as main route and another as backup route. If link bandwidth of all routes is same, we use route that has least number of routers as main route. If the link bandwidth and number of routers are same, we can use any route as main route and another route as backup route.

If you use the following commands to create two static routes for network 30.0.0.0/8, the route will place first route to routing table.

ip route 30.0.0.0 255.0.0.0 20.0.0.210
ip route 30.0.0.0 255.0.0.0 40.0.0.220

If first route fails, the router automatically adds second route to routing table.

Creating, adding, verifying static routes

Routers automatically learn their connected networks. We only need to add routes for the networks that are not available on router's interfaces. For eg, network 10.0.0.0/8, 20.0.0.0/8 and 40.0.0.0/8 are directly connected to Router0. Thus we don't need to configure routes for these networks. Network 30.0.0.0/8 and 50.0.0.0/8 are not available on router 0. We have to create and add routes only for these networks.

Router 0 requirements

- Create two routes for network 30.0.0.0/8 and configure first route (via - Router 1) as main route and second route (via - Router 2) as a backup route.
- Create two routes for host 30.0.0.100 and configure host route as main route and second route as backup route.
- Create two routes for network 50.0.0.0/8 and configure first route as main route and second route as backup route.
- Verify the router adds only main route to routing table.

Router 0 Configuration

```
Router > enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) # ip route 30.0.0.0 255.0.0.0 20.0.0.210
Router (config) # ip route 30.0.0.0 255.0.0.0 40.0.0.220
Router (config) # ip route 30.0.0.100 255.255.255.255 20.0.0.220
Router (config) # ip route 50.0.0.0 255.0.0.0 40.0.0.210
Router (config) # ip route 50.0.0.0 255.0.0.0 20.0.0.220
Router (config) # exit
Router # show ip route static
30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
S 30.0.0.0/8 [10/0] via 40.0.0.25 50.0.0.0/8 [10/0] via
40.0.0.2
```

Access CLI prompt of Router 0 and run following commands.

Router 1 ~~Config~~ requirements

- Create two routes for network 10.0.0.0/8 and configure first route (via - Router 0) as main route and second route as back-up route.
- Create two routes for network 40.0.0.0/8 and configure first route as main route and second route as backup route.

Router 1 configuration

Router > enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) # ip route 10.0.0.0 255.0.0.0 20.0.0.1 10
Router (config) # ip route 10.0.0.0 255.0.0.0 30.0.0.1 20
Router (config) # ip route 40.0.0.0 255.0.0.0 20.0.0.1 10
Router (config) # ip route 40.0.0.0 255.0.0.0 50.0.0.1 20
Router (config) # exit
Router # show ip route static
S 10.0.0.0/8 [10/0] via
20.0.0.1 S 40.0.0.0/8 [10/0] via 20.0.0.1
Router #

Router 2 requirements

Create static routes for network 10.0.0.0/8 and network 30.0.0.0/8 and verify router adds both routes to routing table.

Router 2 configuration

Router>enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config) # ip route 10.0.0.0 255.0.0.0 40.0.0.1
Router (config) # ip route 30.0.0.0 255.0.0.0 50.0.0.2
Router (config) # exit
Router # show ip route static S 10.0.0.0/8 [1/0] via 40.0.0.1
S 30.0.0.0/8 [1/0] via 50.0.0.2
Router #

Verify static routing

on Router 0, we configured two routes for network 30.0.0.0/8 There routes are via Router1 and via Router 2. We set first route as main route and second route as backup route. We can verify configuration in two ways.

By reading ping requests to a PC of network 30.0.0.0/8 and tracing path they take to reach network 30.0.0.0/8. For this you can use tracert command on a PC of network 10.0.0.0/8. The tracert command reads ping requests to destination host and tracks path they take to reach destination.

By listing the routing table entries on router 0. Since a router uses the routing table to forward data packets, you can check routing table to figure out route it, router uses to forward packets for each destination.

We also configured a separate static host route for host 30.0.0.100/8. The router must use this route to forward data packets to host 30.0.0.100/8. To verify this, you can do same testing for host 30.0.0.100/8.

We also configured a backup route for network 30.0.0.0/8. The router must put the backup route to routing table and use it to forward data packets to network 30.0.0.0/8 when main route fails. To verify this, we have to simulate failure of main route.

To simulate failure of main route, you can delete link between Router 0 and Router 1. After deleting the link, do same testing again for network 30.0.0.0/8.

Deleting a static route

To delete a static route, use following steps

- Use 'show ip route static' command to print all static routes.
- Note down the route you want to delete.
- Use the 'no ip route' command to delete the route.

If you have a backup route, the backup route becomes main route when you delete main route.

RESULT:

Thus the Static Routing Configuration is simulated using CISCO Packet Tracer.

From global configuration mode we can enter in interface mode. From there we can configure interface. Following commands will assign IP address on FastEthernet 0/0.

```
Router(config)#interface fastEthernet 0/0
Router(config-if)# ip address 10.0.0.1
255.0.0.0 Router(config-if)# no shutdown
Router(config-if)#exit
Router(config)#
```

interface fastEthernet 0/0 command is used to enter in interface mode.
ip address 10.0.0.1 255.0.0.0 command will assign IP address to interface.
no shutdown command will bring the interface up.
Exit command is used to return in global configuration mode.
Serial interface needs two additional parameters clock and bandwidth. Every serial cable has two DTE and DCE. These parameters are always configured at DCE end. We can use show controllers interface command from privilege mode to check cable's end.

```
Router# show controller serial 0/0/0
Interface Serial0/0/0
Hardware is PowerQUICC MPC860
DCE V.35, clock rate 2000000
[output omitted]
```

Fourth line of output confirms that DCE end of serial cable is attached. If you see DTE here instead of DCE skip these parameters. Now we have necessary information let's assign IP address to serial interface.

15-9-20 EX-11(6) Simulate RIP using CISCO packet Tracer

AIM: To simulate RIP using CISCO packet Tracer

Initial IP configuration

Device	Interface	IP configuration	Connected with
PC0	Fast Ethernet	10.0.0.2/8	Router0's Fa0/1
Router 0	Fa0/1	10.0.0.1/8	PC0's Fast Ethernet
Router 0	S0/0/1	192.168.1.254/30	Router 2's S0/0/1
Router 0	S0/0/0	192.168.1.249/30	Router 1's S0/0/0
Router 1	S0/0/0	192.168.1.250/30	Router 0's S0/0/0
Router 1	S0/0/1	192.168.1.246/30	Router 2's S0/0/0
Router 2	S0/0/0	192.168.1.245/30	Router 1's S0/0/1
Router 2	S0/0/1	192.168.1.253/30	Router 0's S0/0/1
Router 2	Fa0/1	20.0.0.1/30	PC1's Fast Ethernet
PC1	Fast Ethernet	20.0.0.2/30	Router 2's Fa 0/1

Assign IP address to PC0

Double click PC0 and click Desktop menu item and click IP configuration. Assign IP Address referring the above table.

Assign IP address to interfaces of routers

Double click router0 and click CLI and press Enter key to access command prompt of router0.

we need to configure IP address and other parameters on interfaces before we could actually used them for routing. Interface mode is used to assign IP address and other parameters. Interface mode can be accessed from global configuration mode. Following commands are used to access global configuration mode.

Router > enable

Router # configure terminal

Enter configuration commands, one per line. End with CNTL/Z

Router #configure terminal
Enter configuration commands, one per line. End with CNTL/Z
Router(config)# interface serial 0/0/0
Router(config-if)# ip address 192.168.1.249 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# bandwidth 64
Router(config-if)# no shutdown
Router(config-if)# exit
Router(config)# interface serial 0/0/1
Router(config-if)# ip address 192.168.1.253 255.255.255.252
Router(config-if)# clock rate 64000
Router(config-if)# bandwidth
64 Router(config-if)# no
shutdown Router(config-if)# exit

Router # configure terminal - Enter in global configuration mode.
Router(config)# interface serial 0/0/0 - Interface mode.
Router(config-if)# ip address 192.168.1.249 255.255.255.252
- Assign IP address to interface. For serial link we usually use IP address from 30 subnet.
Router(config-if)# clock rate 64000 and Router(config-if)# bandwidth 64
Router(config-if)# no shutdown - interface up
Router(config-if)# exit - Return in global configuration mode

Router 1
Router > enable
Router # configure terminal
Enter configuration commands, one per line. End with CNTL/Z
Router(config)# interface serial 0/0/0
Router(config-if)# no shutdown
Router(config-if)# exit

Router (config)# interface serial 0/0/1
Router(config-if)# ip address 192.168.1.246 255.255.255.252
Router (config-if)# clock rate 64000
Router (config-if)# bandwidth 64
Router (config-if)# no shutdown
Router (config-if)# exit
Use some commands to assign IP address on interface of Router2

Router 2

Router> enable
Router# configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router (config)# interface fast-Ethernet 0/0
Router (config-if)# ip address 192.168.1.245 255.255.255.252
Router (config-if)# no shutdown
Router (config-if)# exit
Router (config)# interface serial 0/0/1
Router (config-if)# ip address 192.168.1.253 255.255.255.252
Router (config-if)# no shutdown

Configure RIP routing protocol

Configuration of RIP protocol is much easier than you think. It requires only two steps to configure RIP routing.

- Enable RIP routing protocol from global configuration mode.
- Tell RIP routing protocol which networks you want to advertise. Let's configure it on Router 0

Router0 (config)# router rip
Router 0 (config-router)# network
10.0.0.0
Router 0 (config-router)# network 192.168.1.252

router rip – Tells router to enable RIP routing
network – allows us to specify networks which we want to
advertise. We only need to specify networks directly
with the router. That's all we need to configure
Follow same steps on remaining routers.

Router 1

Router 1 (config)# router rip
Router 1 (config-router)# network 192.168.1.20
Router 1 (config-router)# network 192.168.1.24

Router 2

Router 2 (config)# router rip
Router 2 (config-router)# network 20.0.0.0
Router 2 (config-router)# network 192.168.1.252
Router 2 (config-router)# network 192.168.1.244

That's it. Our network is ready to take advantage of
RIP routing. To verify setup we will use ping command.
Ping command is used to test connectivity between two
devices. Access command prompt of PC1 and use ping
command to test connectivity from PC0.

RIP protocol automatically manage all routes for us.
If one route goes down, it automatically switches to
another available. To explain this process more clearly
we have added one more route in our network
currently there are two routes between PC0 and PC1.

Route1

PC0 [source/destination -10.0.0.2] <=> Router0 [FastEthernet0/1 - 10.0.0.1] <=> Router0 [serial 0/0/1 -192.168.1.254] <=> router2 [serial 0/0/1 -192.168.1.253] <=> Router2 [FastEthernet0/0 -20.0.0.1] <=> PC1 [destination/source -20.0.0.2]

Route2

PC0 [source/destination - 10.0.0.2] <=> Router0 [FastEthernet 0/1 - 10.0.0.1] <=> Router0 [Serial 0/0/0 -192.168.1.246] <=> Router1 [serial 0/0/0 -192.168.1.250] <=> Router1 [serial 0/0/1 -192.168.1.249] <=> Router2 [serial 0/0/0 - 192.168.1.245] <=> Router2 [FastEthernet 0/0 - 20.0.0.1] <=> PC1 [destination/source -20.0.0.2]

By default RIP will use route that has less hops counts between source and destination. In our network route 1 has, lower hops counts, so it will be selected. we can use tracert command to verify it. Now suppose route 1 is down. we can simulate this situation by removing cable attached between Router0 [S0/0/1] and Router2 [S0/0/1]

RIP will automatically reroute the traffic. Use tracert command again to see magic of dynamic routing.

RESULT:

This RIP is simulated using CISCO Packet Tracer.